

Tacit: A Confidential, Self-Custodial Asset Protocol on Bitcoin

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<https://tacit.finance>

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Abstract

A purely on-chain version of confidential value, anonymous spend, shielded recipient addresses, automated market making, and wrapped Bitcoin would let users hold and exchange programmable assets on Bitcoin L1 without trusting a federation, sidechain, or bridge. Signatures and rangeproofs solve part of the problem; the main benefits are lost if a trusted party still enforces token rules or custodies value. The unlock is a **composition of self-custodial primitives**: two Groth16 circuit families (anonymous unique-spend and amount confidentiality), a Runes-style indexer-validated meta-protocol, and a fixed-supply tacit asset (TAC). TAC's market value lets indexer-validated state enforce economic constraints, and its intrinsic privacy value as a confidential, self-custodial asset justifies its existence alongside Bitcoin. Any indexer running the specification reaches the same verdict from chain data alone. Wrapped Bitcoin is locked cryptographically at a Taproot key derived from a mixer note's own secret: one leaf, two locks. Amount-granular fungible BTC instead replaces the federation with an LP-share lien on the canonical AMM pool, with TAC economic over-collateralization as the bond. Together these primitives compose into real Bitcoin DeFi as a self-custodial layer — confidential assets, native AMM, yield farms, atomic-OTC marketplace, wrapped BTC — anchored to Bitcoin L1, recoverable from each user's private key, and free of federation, custodian, or operator-set trust. The cryptographic claims hold under standard assumptions; the economic claims hold conditional on TAC market depth and an active liquidator population, both enumerated explicitly.

1 Introduction

Bitcoin solved double-spend over a peer-to-peer network. Doing more than moving sats — confidential transfers, anonymous spend, automated market making, wrapping Bitcoin into programmable form — has historically required either off-chain proof exchange (RGB, Taproot Assets), a federated sidechain (Liquid), a rollup or sidechain with an operator set (Citrea, Botanix), or a custodial or threshold-bonded wrapper (WBTC, tBTC). Each reintroduces a trusted party.

What is needed is a layer that (i) keeps every byte of state on Bitcoin L1, (ii) hides amounts cryptographically, (iii) supports anonymous spend without a coordinator, (iv) lets Bitcoin be wrapped into a fungible asset without a federation, and (v) recovers entirely from a private key. Tacit is that layer — a composition of self-custodial primitives that delivers DeFi on Bitcoin L1 without a runtime, custodian, or operator set.

2 Indexer-validated meta-protocol

Tacit is an indexer-validated layer on Bitcoin L1. Envelopes ride inside Bitcoin Taproot script-path witnesses; Bitcoin nodes do not interpret them. *Indexers* — open-source clients anyone can run — scan confirmed transactions and reach the same verdict from chain data alone. This is the trust model Runes and Ordinals already use to underwrite market value¹: two indexers seeing the same byte sequence reach the same accept/reject decision, byte-for-byte. There is no voting, no quorum, no leader. Determinism is the substrate.

The trust target is therefore not “indexer determinism in the abstract” but the **published specification and its open-source reference implementations** — the same shape Bitcoin Core has at L1. Contested edge cases resolve via SPEC clarification and cross-implementation test vectors, not by quorum vote. This is a softer property than a federation and a stronger one than off-chain proof distribution, but it is not a solved problem in the abstract: it depends on the protocol’s open-source process remaining contestable. §13 enumerates the limits this implies.

The contribution is to use that substrate for more than tokens. The same consensus-of-indexers that enforces who-owns-what can also enforce who-bonded-what and who-equivocated-when, *provided* there exists a tacit asset the protocol can natively slash. TAC is that asset: a fair-launched tacit asset with a fixed 21,000,000-base-unit cap², deployed via the same permissionless T_PETCH/T_PMINT mechanism any other auditable-supply token uses. The protocol uses TAC as the bond layer for cBTC.tac (§6), workers (§9), and bounded governance.

Indexer determinism + a fixed-cap tacit asset together replace both the federation and the smart-contract VM. Sidechains keep the federation (Liquid, ~11-of-15 multisig); rollups and Bitcoin L2s keep a VM but inherit an operator set (Citrea, Botanix); each reintroduces a trust party tacit avoids. Where cryptography can enforce a property directly — custody of BTC at L1, amount confidentiality, anonymous spend — the protocol uses cryptography. Where it cannot — fungibility of wrapped BTC, soft-confirms before Bitcoin confirmation — TAC stake fills the gap. Because the cap is fixed and slashed TAC burns against it, the bonded layer’s strength compounds: each burn permanently shrinks the circulating float against a fixed cap, so every remaining holder’s relative share strengthens and every future bond stacks against a smaller supply. Slashing schemes that recycle proceeds into recoverable treasuries, or that coexist with continuous issuance, do not have this monotone property.

3 Confidential value

Every on-chain amount is a Pedersen commitment on secp256k1³:

$$C = a \cdot H + r \cdot G,$$

where G is Bitcoin’s generator, H is a NUMS generator (no known $\log_G H$), a is the amount, and r is a blinding scalar. Commitments are perfectly hiding and additively homomorphic, so supply conservation holds across a transaction by checking

$$\sum_{\text{out}} C - \sum_{\text{in}} C = r_{\text{ex}} \cdot G,$$

¹Rodarmor, C. *Runes Protocol*. 2024.

²SPEC-WORKER-BOND-AMENDMENT — *Why burn the slashed TAC instead of pooling it*: TAC fixed cap, burn-on-slash semantics.

³Pedersen, T. P. *Non-interactive and information-theoretic secure verifiable secret sharing*. CRYPTO 1991.

where r_{ex} is the excess. A BIP-340 Schnorr signature⁴ under $r_{\text{ex}} \cdot G$ — a Mumblewimble-style **kernel signature**⁵ — proves $\sum_{\text{out}} a = \sum_{\text{in}} a$ without revealing any a . Range is enforced by one aggregated Bulletproof⁶ covering all outputs at $n = 64$ bits, $m \in \{1, 2, 4, 8\}$.

The recipient’s blinding and amount-keystream are both derived from the ECDH shared secret of the sender’s private key and the recipient’s public key, with domain-separated HMAC-SHA256. The sender’s pubkey appears in the input witness; the recipient re-derives every credit from their private key and chain alone. **No share-link, no sync server.**

4 Anonymous unique-spend

Amount privacy alone leaves the address graph visible. Tacit’s mixer adds a Tornado-style shielded pool per $(\text{asset_id}, \text{denom})$ pair. Deposit appends a Poseidon-hash leaf⁷ $\text{leaf} = \text{Poseidon}_3(s, \nu, \text{denom})$ to a depth-20 Merkle tree the indexer maintains. Withdrawal produces a Groth16 proof over BN254⁸ of the statement

$$\exists (s, \nu, \pi) : \text{leaf}(s, \nu, d) \in \text{root}, \text{Poseidon}_1(\nu) = \text{null}, r_{\text{leaf}} = \text{Poseidon}_2(s, \nu),$$

with public inputs $(\text{root}, \text{null}, d, r_{\text{leaf}}, \text{bind})$, where bind is SHA-256 over withdrawal-specific data $(\text{asset id}, \text{denomination}, \text{nullifier}, \text{recipient commitment}, r_{\text{leaf}})$ reduced to a BN254 Fr element. The validator recomputes it externally; because the circuit constrains bind as a public input, Groth16’s input-commitment vector binds the proof to that specific value, and any reuse against a different recomputed tuple fails verification — preventing replay or front-running. Indexers reject any reused null. The deposit and withdrawal are unlinkable inside the anonymity set of unspent leaves. The same circuit underpins cBTC.zk slot operations (§5), where leaf-level privacy holds but each slot’s backing BTC UTXO is still a visible Bitcoin chain-graph element — protocol-layer privacy without BTC-layer privacy⁹.

5 cBTC.zk: one leaf, two locks

BN254 Fr scalars (order $\approx 2^{254}$) embed into secp256k1’s scalar field (order $\approx 2^{256}$) without reduction, so $r_{\text{leaf}} \cdot G_{\text{secp256k1}}$ is a valid Bitcoin public key. We lock real BTC at a key-path Taproot output

$$K_{\text{btc}} = r_{\text{leaf}} \cdot G_{\text{secp256k1}} = C - d \cdot H,$$

derivable by any indexer from the on-chain Pedersen commitment C . **The same scalar that proves set membership inside the SNARK is the spending key of the backing UTXO.** A user who can withdraw the note can spend the BTC; nobody else can. No federation, no co-signer, no oracle, no escape script — one leaf, two locks. We call this asset **cBTC.zk**.

⁴Wuille, P.; Nick, J.; Ruffing, T. *BIP-340: Schnorr signatures for secp256k1*. 2020.

⁵Jedusor, T. E. *Mumblewimble*. 2016; Poelstra, A. *Mumblewimble*. 2016.

⁶Bünz, B.; Bootle, J.; Boneh, D.; Poelstra, A.; Wuille, P.; Maxwell, G. *Bulletproofs: Short proofs for confidential transactions and more*. IEEE S&P 2018.

⁷Grassi, L.; Khovratovich, D.; Rechberger, C.; Roy, A.; Schofnegger, M. *Poseidon: A new hash function for zero-knowledge proof systems*. USENIX Security 2021.

⁸Groth, J. *On the size of pairing-based non-interactive arguments*. EUROCRYPT 2016.

⁹See `spec/GLOSSARY.md` — *Two privacy layers, not one*.

Transfer rekeys atomically. Moving a cBTC.zk slot to a new holder spends the old K_{btc} and creates a new $K'_{\text{btc}} = r'_{\text{leaf}} \cdot G_{\text{secp256k1}}$ in the same Bitcoin transaction (T_SLOT_ROTATE, T_SLOT_SPLIT, or T_SLOT_MERGE). The new key is derived from a secret the recipient controls (the sender never learns it), so the previous holder retains no spending authority over the backing BTC UTXO. cBTC.zk is therefore transferable as a first-class slot asset, not merely a self-held vault note.

Non-upgradability is the price of trustlessness. The slot layer has no escape script and no break-glass. The same cryptography that excludes federations from custody also excludes recovery if a circuit bug is ever found. A successor pool with a new circuit ships as a separate (*asset_id*, denom) surface; existing slots stay on their original key and migrate slot-by-slot through the standard withdraw-and-redeposit flow. This is the same trade Bitcoin itself makes at L1 — cryptographic rigidity in exchange for the absence of an upgrade party.

Any spend of K_{btc} without a matching T_SLOT_BURN envelope is observable from chain alone (the slot’s nullifier is missing from the consumed set). This SLASH_DETECTED event is the primitive on which later economic constructions rest.

6 cBTC.tac: bonded fungibility

cBTC.zk is unit-granular at fixed denominations and trustless. DeFi expects amount-granular fungible assets. **cBTC.tac** composes a cBTC.zk slot with an LP-share lien on the canonical (cBTC.zk, TAC) AMM pool. Minting requires the depositor to LP both sides; the indexer records a lien on the resulting LP-share UTXO and mints fungible cBTC.tac equal to the slot’s BTC value. The lien is enforced by the wallet’s universal commitmentForUtxo resolver: any unauthorized spend is treated as nonexistent, so AMM swaps, transfers, and farms refuse to consume liened collateral¹⁰.

If the depositor rugs the underlying slot (SLASH_DETECTED), their TAC contribution moves to a shared insurance pool that compensates all outstanding cBTC.tac holders pro rata. With initial bond ratio ≥ 2.0 relative to slot BTC value and force-close defending at $1.2\times$, the rug is negative expected value while the liquidator population operates as designed. **cBTC.tac is trustless on the BTC anchor side (cryptographic) and over-collateralized on the fungibility side (economic) — the same shape as DAI but with no smart-contract runtime.** This makes cBTC.tac a first-class tacit asset: it transfers via standard CXFER, swaps in any pool, deposits into the mixer, and stakes into yield farms. Bitcoin gets a fungible BTC-denominated asset inside a DeFi stack anchored to L1, without a custodian.

Stress behavior and reflexive collateral. The construction is procyclical. TAC’s market value derives in part from its role as the bond, so a stress event that strains cBTC.tac can also strain TAC — the collateral is partially **reflexive**, not merely correlated. The $2.0\times$ initial cushion, the permissionless force-close path triggered when LP-share BTC value falls below $1.2\times$ the slot, and the shared insurance pool bound the damage. Non-bond TAC demand (AMM market-making, payments, governance) is what mitigates the reflexivity in practice. The trust model under stress is “rational liquidators arrive in time,” the same shape Maker has. Maker collateralizes against ETH’s mature exogenous demand; TAC’s parallel demand develops through the non-bond uses enumerated in §13 — worker bonding, governance, AMM denomination, protocol-fee revenue (§8), farm rewards, and TAC’s role as a native confidential asset.

¹⁰SPEC-CBTC-TAC-AMENDMENT §5.47 — lien model (trustless collateral without covenants).

7 Three privacy capabilities, composable and optional

Privacy is exposed as three orthogonal capabilities, each opt-in at a different granularity. Users pick the posture that fits their use case instead of forcing one stance on everyone.

Capability	Default	Technique	What it hides
Shielded amount	On (every transfer)	Pedersen commit $C = aH + rG$, aggregated Bulletproof for $a \in [0, 2^{64})$, kernel signature on the excess; ECDH-derived recipient keystream	Per-output amounts. Only the public <code>burned_amount</code> field of <code>T_BURN</code> is exposed by design — for auditability of supply reductions.
Shielded address	Opt-in per receipt	Blinded-pubkey commit $= P_{\text{recipient}} + b \cdot G$ with $b = \text{HMAC}(\text{shared} \parallel \text{anchor})$, $\text{shared} = \text{ECDH}(\cdot)$. Static handle <code>tcs1...</code> encodes the recipient pubkey opaquely	The recipient's on-chain identity. Each receipt lands at a per-transaction unique <code>P2WPKH(hash160(commit))</code> with no apparent link to the published handle. Same key-tweak primitive as BIP-341 ¹¹ / BIP-352 ¹² ; no new ceremony.
Mixer pool	Opt-in UTXO	Tornado-style Poseidon-Merkle commitment with Groth16 proof of unspent-leaf membership and nullifier reveal (§4)	The link between deposit and withdraw. Inside the pool's anonymity set, which leaf was touched is hidden; both endpoints belong to the same (<code>asset_id</code> , <code>denom</code>) surface.

The three layers are orthogonal because each operates at a different position in the wire: **shielded amount** on the commitment value, **shielded address** on the output script, **mixer pool** on the protocol-tree leaf. A given UTXO can carry any subset independently, and a wallet can dial privacy up or down per-payment without giving up recovery.

Practical postures. A *merchant* keeps default shielded amounts and a static public identity for accounting — balances are private but counterparties remain identifiable for invoicing and reconciliation. *Payroll* runs a batched CXFER from one treasury key to N employees, each of whom publishes a `tcs1...` shielded address: every employee's on-chain pay line lands at a fresh, unlinkable marker, and the employee re-derives the tweaked signing key from their wallet seed alone. A *privacy-conscious user* routes outbound flow through the mixer pool and receives inbound at a shielded address — unlinkable at both endpoints, with full unlinkability inside the chosen anonymity set. Each posture composes **the same three primitives** differently; the protocol itself does not change between them.

8 Native AMM and LP-bonded farms

The AMM follows the same virtual-pool pattern as the mixer: a pool of (R_A, R_B, S) is **just numbers** that the indexer reconstructs from chain. Two trader paths:

- `T_SWAP_VAR` — per-trade against the spot curve $\Delta_{\text{out}} = R_B \gamma \Delta / (R_A \gamma_d + \gamma \Delta)$ (γ and γ_d are the per-pool fee numerator and denominator, e.g. $\gamma = 997$, $\gamma_d = 1000$ for a 30 bps

pool) using only Pedersen + Bulletproof + kernel sig. **No Groth16, no ceremony, and per-trade Δ is cleartext on chain** — the explicit trade-off for skipping the SNARK. Variable-amount $[Y, X]$ semantics, immediate settlement. Privacy-sensitive flow routes through T_SWAP_BATCH instead.

- T_SWAP_BATCH — uniform-clearing-price batch¹³ over N hidden per-trader amounts, settled by one Groth16 proof. Intra-batch sandwich attacks are structurally impossible because every trader in a batch clears at the same P_{clear} .

LP shares are themselves confidential tacit assets, so they compose with the mixer (anonymous LP positions) and with cBTC.tac (LP-share-lien collateral). Yield farms (T_LP_BOND) stream reward emissions to bonded LP shares without a smart contract: the farm treasury is a virtual indexer-attested quantity; bond receipts are P2WPKH dust markers keyed to per-bond accrual snapshots; the launcher’s reward asset enters via kernel-sig closure and exits when the validator credits a fresh reward output at unbond¹⁴. Bootstrap depth is incentivized without an on-chain treasury or custody role.

Optional protocol-fee skim. Pools may pin a fee skim at POOL_INIT, directing a portion of trade revenue to a configurable recipient. Canonical pools direct it to the TAC treasury, so AMM volume accrues structural revenue to TAC holders alongside organic LP fees. **TAC is the protocol’s DAO-governed DeFi-native unit** — canonical to the AMM, bonding, and governance from launch, not a generic asset adapted to collateral after the fact. The same trade activity that prices cBTC.tac through TWAP collateralization also accrues value at the bond layer. With the full DeFi primitive set on one stack — fungible wrapped BTC, AMM, LP yield, farm incentives, snapshot airdrops, future stablecoin variants — pool counterparties span TAC, cBTC.tac, and any other tacit asset, so depth compounds within the protocol’s value-accrual surface rather than fragmenting to external venues. A structural difference from MakerDAO, where the AMM that prices ETH/DAI runs on external venues and its fees flow elsewhere.

9 Validator honesty and bonded diversity

The channel layer (T_INTENT_ATTEST) lets workers offer ~ 30 s soft-confirmations on pending intents — pre-broadcast trade commitments that haven’t yet settled on Bitcoin. Limit orders that never match cost zero on-chain; matched orders amortize into batched settlement. The construct is only useful if equivocation is costly.

A worker who wants traders to honor its attestations posts a TAC bond via T_WORKER_BOND_OPEN. The bond UTXO is an ordinary TAC-bearing UTXO carrying an **indexer-recorded lien** — no covenant needed; tacit-aware wallets refuse to consume liened TAC. Any party who observes two attestations signed by the same `worker_pubkey` at the same $(scope_id, height)$ with conflicting `intent_pool_hash` can submit them as evidence; the indexer reattributes a configurable bounty fraction to the reporter and **burns the remainder**, contracting circulating TAC against its fixed 21M-base-unit cap. Any observer has positive-EV incentive to report. The structure mirrors Ethereum’s PoS slashing¹⁵ but is enforced by indexer consensus rather than a smart contract, and inherits the monotone-deterrent property from §2.

¹³Walras, L. *Éléments d’économie politique pure*. 1874; Gnosis Protocol uniform-clearing-price batches, 2019; Penumbra ZSwap, 2023.

¹⁴SPEC-AMM-FARM-AMENDMENT — virtual-treasury MasterChef-style farms.

¹⁵Buterin, V., et al. *Ethereum Proof-of-Stake / Casper FFG*. 2020.

Workers cannot make invalid envelopes valid; honest competition for intent flow plus burn-on-equivocation suffices to align worker behavior. Anyone can run a worker, and the dapp’s **tacit-mesh** cross-checker turns that diversity into evidence: a worker that quietly omits an intent from its published list while signing an attestation committing to the omitted set is caught by a SHA-256 comparison across ≥ 2 subscribed workers. The mesh produces evidence; the bond turns it into a slash.

Bond-class accounting. Worker bonds and cBTC.tac collateral share the same TAC float and are both recorded as indexer liens against the fixed cap. A system-wide governance parameter caps the bonded fraction of circulating TAC, and burns from either class contract supply for the other. The classes compete for float but compose cleanly: one TAC, one cap, one ledger of liens.

10 Self-custody and recovery

A wallet recovers full state from its private key and the Bitcoin chain alone. The derivation paths by primitive:

- *Confidential transfers:* trial-decrypt outputs via ECDH against the sender pubkey carried in the input witness.
- *Own etches and mints:* re-derive supply blinding/keystream from the commit-tx anchor and the wallet’s own key.
- *Permissionless mints and mixer withdrawals:* the opening (a, r) is published cleartext in the envelope (verified via the Pedersen equation).
- *AMM and farm positions:* the public amount $(\text{share_amount}, \Delta_a, \Delta_b)$ sits in the envelope; the chain commitment uses a deterministic blinding derived from the wallet key plus a per-tx anchor.
- *Shielded-address receipts:* scan a bounded **xferseen** index per asset and re-derive the per-tx tweak b from ECDH and the anchor; the spending key is $sk_{\text{recipient}} + b \bmod n$.

Shielded-pool deposits and cBTC.zk slots. Each deposit’s (secret, ν) pair is wallet-derived at deposit time and consumed at withdraw time. Standard tacit wallets derive the pair deterministically from the wallet seed (a BIP-32-style sub-key seeds the per-deposit CSPRNG, anchored on the deposit tx), so recovery re-derives candidate notes and scans the pool’s leaf set for matches — privkey-only, same as the rest of the protocol. The protocol consumes any valid (secret, ν) pair without distinguishing how it was generated, so a wallet may also choose per-deposit CSPRNG with out-of-band note backup (the Tornado / Privacy Pools posture); both paths produce identical on-chain leaves and identical withdraws. **Fungible cBTC.tac balances** held as ordinary tacit-asset UTXOs recover from the wallet key alone like any other tacit asset.

A user who loses every device and every backup except their private key reconstructs every asset class from the chain alone — the property native Bitcoin already has, extended to a full confidential DeFi stack.

11 Fee-aware cryptography

Witness bytes are the dominant cost. The protocol economizes by picking the lightest primitive that proves the necessary property:

- **Bulletproofs+** replaces Bulletproofs for CXFER and AXFER — same Pedersen, same kernel sig, $\sim 14\%$ smaller witness across every transfer¹⁶.
- **Sigma cross-curve binding** (169 bytes, microseconds) links a secp256k1 chain commitment to a BabyJubJub in-circuit commitment, avoiding the $\sim 10^6$ -constraint cost of doing secp256k1 EC math inside a BN254 SNARK.
- **T_SWAP_VAR carries no Groth16**. Variable-amount fills against a known curve are checkable from public data; ceremonies are reserved for the surfaces where set-membership privacy or multi-party amount confidentiality is load-bearing.
- **Multi-hop routing** (T_SWAP_ROUTE) and **batched preauth-take** atomically settle N legs in one Bitcoin tx, amortizing the ~ 10 KB witness floor across many fills.
- **Lazy accrual** of LP and farm rewards uses a fixed-point per-share accumulator (Q.96), so reward crystallization is $O(1)$ per state-touching event rather than $O(LP)$ s, matching MasterChef¹⁷.

The composition gives smart-contract-shaped properties — AMM trading, collateralized wrapping, batched settlement, programmatic yield farms — delivered without a VM, without a sidechain, without leaving Bitcoin L1.

12 The protocol as one ecosystem

The primitives fit together as a single system. Any tacit asset works in any primitive uniformly, so the surfaces below compose into one self-consistent stack rather than a menu of disconnected products.

- **Fair-launched native asset**. TAC, 21M-base-unit cap, deployed via permissionless T_PETCH / T_PMINT. Wire-format-wise a regular tacit asset (amount-private, shielded-address-compatible, mixer-composable); structurally canonical to the AMM, bonding, and governance from launch.
- **Asset-creation arm**. CETCH issues confidential-supply assets (optionally mintable); T_PETCH + T_PMINT covers fair-launch capped issuance with publicly auditable cumulative supply and a height window; T_DROP + T_DCLAIM distributes existing supply via snapshot-based merkle claims with optional ETH-signature gating.
- **Confidential AMM and yield farms**. Constant-product pools between any two tacit assets, two trader paths (T_SWAP_VAR cleartext-amount, T_SWAP_BATCH Groth16-private uniform clearing), T_LP_BOND / T_LP_HARVEST farm-reward layer over virtual treasuries.
- **Cryptographic BTC slots and collateralized fungible BTC**. cBTC.zk for cryptographic 1:1 wrapping (fixed-denomination slots), cBTC.tac for fungible BTC-denominated

¹⁶Chung, H., et al. *Bulletproofs+*. 2020.

¹⁷SushiSwap. *MasterChef contract*. 2020.

DeFi (amount-granular, TAC-collateralized) — both first-class tacit assets that transfer, swap, mix, and farm like any other.

- **Atomic OTC marketplace.** `T_AXFER`, `T_AXFER_VAR`, `T_SWAP_ROUTE`, batched preauth-take: single-tx confidential-asset-vs-BTC settlement with maker/taker pre-signing that prevents either side from griefing.
- **Bonded soft-confirm layer.** `T_INTENT_ATTEST` preconfirmations, `T_WORKER_SLASH` equivocation slashing on cryptographic evidence, tacit-mesh cross-worker evidence production.
- **Bounded governance.** `T_GOV_*` opcodes for TAC-weighted parameter votes within explicit safety bands; load-bearing mechanics remain immutable without a formal amendment.

Smart-contract-shaped without a VM. Smart contracts give Ethereum users programmable assets, automated market making, collateralized issuance, atomic multi-leg settlement, yield distribution, governance, and snapshot-based airdrops. Tacit delivers the same outcomes by composing the primitives above. Each opcode is a deterministic state-transition rule any indexer reaches the same verdict on; the trade-off is real (no arbitrary user-written logic) and the gains are commensurate (no VM attack surface, no operator set, no L1 change to add new primitives).

Smart-contract pattern	Tacit primitive(s)
ERC-20 transfer + optional mint/burn	<code>CETCH</code> + <code>CXFER</code> + <code>T_MINT</code> + <code>T_BURN</code> (Pedersen-committed amounts; mintable if elected at etch; burn amount is public for auditability).
Fair-launch capped permissionless mint	<code>T_PETCH</code> + <code>T_PMINT</code> + <code>T_BURN</code> (publicly auditable cap, height window, depth-3 reorg gate).
Uniswap V2 AMM	<code>T_LP_ADD</code> / <code>T_LP_REMOVE</code> / <code>T_SWAP_VAR</code> / <code>T_SWAP_BATCH</code> (virtual pool, optional per-trader amount confidentiality).
MakerDAO CDP	<code>cBTC.tac</code> via <code>T_CBTC_TAC_DEPOSIT</code> (slot + TAC lien, 2× cushion, 1.2× force-close, shared insurance pool).
Tornado Cash privacy pool	<code>T_DEPOSIT</code> / <code>T_WITHDRAW</code> (Poseidon-Merkle + Groth16 + nullifier set).
MasterChef yield farm	<code>T_FARM_INIT</code> / <code>T_LP_BOND</code> / <code>T_LP_HARVEST</code> (virtual treasury, Q.96 lazy accrual).
Merkle Distributor airdrop	<code>T_DROP</code> / <code>T_DCLAIM</code> (Merkle-snapshot eligibility, per-claim cap, expiry height).
1inch-style multi-leg router	<code>T_SWAP_ROUTE</code> (atomic 2–4-hop AMM routing in one Bitcoin tx).
Optimistic-rollup fraud proof	<code>T_WORKER_SLASH</code> (cryptographic two-attestation evidence, burn against fixed cap).
Compound/Aragon governance	<code>T_GOV_PROPOSAL</code> / <code>T_GOV_VOTE</code> / <code>T_GOV_VETO</code> / <code>T_GOV_EXECUTE</code> (TAC-weighted, safety-band-bounded).

Primitives are immutable; composition is uniform. Each row above is an immutable state machine fixed at SPEC time (or governance-bounded within explicit safety bands). Users compose primitives; they cannot redefine them. Any tacit asset — `CETCH`’d tokens, LP shares, `cBTC.tac`, TAC — transfers via `CXFER`, swaps in any pool, deposits into the mixer, and bonds into farms or worker positions. This is the ERC-20-shape composability that gave Ethereum DeFi its compounding effect, ported to Bitcoin L1 without a VM.

The stack strengthens with use. Adoption on any axis lifts the others: wider TAC use increases shielded-transfer volume and mixer anonymity sets; deeper AMM liquidity tightens cBTC.tac’s TWAP-based collateral signal and widens the channel back to plain BTC; a larger bonded base against the fixed cap raises the deterrent for every future bond; richer farm emissions deepen LP positions that feed the same liquidity. The same indexer determinism and the same fixed-cap asset underwrite all of it, and the same primitives compose across all of it. The ecosystem reinforces itself by being used.

What this misses: arbitrary user-written contracts. A new primitive requires a new opcode (and a new circuit if cryptography is load-bearing), shipped as an additive opcode under SPEC §5.5. The bar is higher than deploying a smart contract — but the surface area subject to bugs is the protocol specification, not every user-written script, and the soundness argument shrinks accordingly. Tacit is an algebra of primitives, not an execution environment.

13 Trust model and limits

The protocol’s load-bearing assumptions deserve naming.

Indexer reference implementation. The trust target at the indexer layer is the published specification and the open-source reference implementations that converge on it, the same shape Bitcoin Core has at L1. Two indexers running the same code over the same bytes reach the same verdict; achieving *the same code* is a social process. Contested edge cases (which have occurred for Runes / Ordinals indexers in production) resolve via SPEC clarification and cross-implementation test vectors, not by quorum vote. After the beta phase stabilizes the wire format under market testing, the TAC DAO (§12) takes on bounded stewardship of future protocol versions and reference-implementation maintenance, under the same safety-band governance that scopes all other parameter changes. Load-bearing mechanics — conservation, slashing semantics, cryptographic primitives — remain immutable without a formal SPEC amendment.

TAC bootstrap and market reality. TAC’s initial distribution went via airdrop to members of the Ethereum-rooted zOrg DeFi DAO — a protocol that has operated on Ethereum for over a year — with snapshot CSVs preserved at `airdrop/` in the repo. TAC shares have circulated on the open DeFi market since the airdrop. The distribution carried real economic cost on both sides: zOrg eligibility was earned through ETH spending (share purchase, protocol fees, or LP farming on Ethereum), and airdrop fulfillment paid Bitcoin fees to broadcast the on-chain envelopes. On Bitcoin, TAC is etched via the public-mint `T_PETCH` / `T_PMINT` mechanism with a fixed 21M-base-unit cap auditable from chain alone. An active OTC market for TAC operates on the protocol itself — settling asset-vs-BTC trades atomically at `tacit.finance` through `T_AXFER` / `T_AXFER_VAR` and the atomic-intent flow — so market liquidity is already in place, not theoretical.

Persistent utility and reflexive collateral. The protocol’s security gains weight as TAC market depth grows, and the protocol structures multiple persistent uses to support that growth: bonding workers against equivocation, weighting bounded governance, denominating the canonical (cBTC.zk, TAC) and (cBTC.tac, TAC) AMM pools that route every BTC-paired trade, receiving protocol-fee revenue from those pools (§8), paying out yield-farm rewards, and circulating as a confidential native asset with full shielded-amount and shielded-address support. The collateral is partially **reflexive** — a stress event that strains cBTC.tac can also strain TAC — and the 2.0×

cushion, $1.2\times$ force-close, and shared insurance pool bound the damage. The non-bond uses above mitigate the reflexivity by spreading TAC’s value across multiple correlated demand sources; only its open-market trading and standalone confidential-asset utility are truly exogenous to cBTC.tac stress.

Network effects on privacy. TAC’s confidential-asset properties give adoption a self-reinforcing component at every privacy layer, and the protocol enters this loop with an active wallet base in the thousands already transacting through TAC’s shielded primitives. Every TAC transfer is amount-hidden by default and any receipt can land at a per-transaction-unique shielded-address marker, so wider adoption directly increases the volume of shielded activity on chain — more cover traffic for an observer to sift through. Mixer use compounds this further: anonymity sets in TAC mixer pools grow as a fraction of holders route through the shielded path. Deeper liquidity in the canonical (cBTC.zk, TAC) and (cBTC.tac, TAC) pools widens the channel between confidential balances and plain BTC sats — exits to Bitcoin clear larger sums with less price impact and, when routed through the mixer, with stronger unlinkability. cBTC.tac’s TWAP-based collateral check benefits from the same depth. Economic-security gains and privacy gains move together at every layer of the stack.

Bitcoin L1 fee regime. A confidential transfer carries ~ 10 KB witness, settling at $\sim 2,500$ – $3,000$ vBytes after the SegWit discount. Tacit is a high-value-transfer instrument, not a low-value payments rail. Bulletproofs+ ($\sim 14\%$ smaller) and multi-leg routing (T_SWAP_ROUTE, batched preauth-take) amortize the cost across more fills, and Bitcoin covenant upgrades (§14) compress further, but no design choice eliminates the L1 witness floor.

Reorg discipline. Per-pool deposits, fair-launch mints, and AMM operations credit only at Bitcoin confirmation depth ≥ 3 — a transient L1 spend that gets reorged out is rolled back along with its tacit-asset effect. cBTC.zk slot spends are confirmed by Bitcoin itself; the indexer-tracked nullifier set rolls back with the chain, so the same r_{leaf} is re-usable after a reorg iff the corresponding L1 spend also reorged out. Worker slashes follow the same rule: an evidence transaction reorged out reverses the slash, and the dapp’s mesh continues to surface the original conflict regardless.

Cross-batch curation MEV. T_SWAP_BATCH’s uniform clearing price defeats *intra-batch* sandwich attacks by construction. Cross-batch exclusion — a settler omitting intents from a batch they assemble — is currently bounded by tip economics, T_INTENT_ATTEST preconf-equivocation evidence, and arbitrage realignment in the next batch. The reserved T_EXCLUSION_CLAIM opcode (drafted) closes this surface in a follow-up amendment via consensus-level slashing of demonstrable exclusion.

Ceremony surface. The mixer’s `withdraw.circom` runs on one global Phase 2 ceremony (canonical bundle content-addressed and hardcoded in the dapp); cBTC.zk reuses it without modification. The AMM’s three sub-circuits share Phase 1 (`pot18`) and run independent Phase 2 chains anchored to one Bitcoin-block beacon at finalization — a single ceremony for the AMM, not a per-pool one. The proof system’s zero-knowledge property protects witness privacy unconditionally; soundness depends on ≥ 1 honest contributor across the ceremony’s participants.

Composable compliance. The mixer pool composes with association-set techniques (the Privacy Pools construction for opt-in transparency to known-good UTXO sets) without modifying the

protocol — an association-set indexer is an additive client-side filter on the same on-chain envelope set. Tacit ships the cryptographic primitive; downstream operators choose the policy.

Threat-model summary. In scope: indexer-spec correctness (auditable and reproducible from chain), Bitcoin consensus, Groth16 / Pedersen / Schnorr soundness under standard assumptions, and rational worker and liquidator behavior. Out of scope: state-level adversaries that compromise Bitcoin itself, dominant reference-implementation collusion that violates the SPEC, and tail-correlated TAC/BTC collapse beyond the bounded-defense surface above. The cryptographic claims hold under standard assumptions; the economic claims hold conditional on TAC market depth and an active liquidator population.

14 Upgrade path

Tacit is built to absorb improvements in three independent dimensions: new Bitcoin opcodes, new cryptography, and adjacent privacy protocols. Each new tacit opcode is purely additive: indexers that don't yet recognize it skip the envelope; indexers that do recognize it apply the new rule (SPEC §5.5 unknown-opcode rule). Upgrades land without breaking existing state or requiring users to migrate value.

Bitcoin covenants. A covenant-style primitive — a CTV-style template check, an OP_CAT-enabled introspection construction, an OP_VAULT-successor design, or a TLUV-style equivalent — activates several already-reserved opcodes and tightens existing ones:

- **Trustless fractional cBTC.zk** (T_SLOT_FRACTIONALIZE / T_SLOT_RECONSOLIDATE, opcodes 0x4D / 0x4E, drafted) splits a cBTC.zk slot into fungible shares and recombines them without a TAC bond. The cryptographic spec is complete; only the Bitcoin-side enforcement is missing. cBTC.tac remains for users who prefer the economic-collateral path; the fractional opcodes become the structurally-trustless fungible BTC path.
- **Aggregated cBTC.zk mixing.** Multiple slots share one Bitcoin UTXO, giving cBTC.zk slot operations real BTC-chain-graph privacy on top of the protocol-tree privacy they already have (§4).
- **On-chain bid escrow.** Today's bid layer is off-chain because Bitcoin script cannot bind a buyer's pre-signed sats input to an unknown future asset outpost; a CTV-style template check closes that gap, removing one of the last off-chain trust surfaces.
- **Covenant-restricted swap inputs** close the residual intent-race window that currently relies on ~3-block depth gating in the AMM.

ZK in Bitcoin script. Recent work on Shielded CSV¹⁸ and OP_CAT-enabled SNARK verifiers would let some validation move from the indexer to Bitcoin consensus itself. Tacit's Groth16 stack is already aligned: `withdraw.circom`'s verifying key is content-addressed and ceremony-anchored, so a proof a worker accepts today verifies under a script-level checker tomorrow without re-running the ceremony.

¹⁸Nick et al., 2024, "Shielded CSV: Private and Efficient Client-Side Validation," and the broader OP_CAT-enabled SNARK verifier line of research.

Schnorr improvements. MuSig2 / FROST give multisig mint-authority with no wire-format change — the `mint_authority` field is just a 32-byte x-only pubkey. Cross-Input Signature Aggregation (CISA), if it lands at consensus, would shrink batched settlements where each fill carries its own signature today.

Interim privacy composition. Before covenants land, two neighboring techniques compose with tacit’s existing surfaces:

- **CoinJoin upstream of mixer deposits or cBTC.zk mints.** A Bitcoin-layer CoinJoin breaks the fee-source-UTXO clustering that operationally weakens self-mix unlinkability — a wallet that funds its deposit from a CoinJoin output has no chain-graph trail back to its previous identity.
- **BIP-352 silent payments for the plain-BTC funding leg.** Silent payments compose orthogonally with the shielded address: tacit hides the recipient marker on tacit-asset receipts; BIP-352 hides it on plain-sats receipts. Tacit defers a tacit-flavored stealth-sats scheme rather than competing with the BIP-352 standard, and uses the mixer pool for the same use case until BIP-352 wallet adoption converges.

Why additivity holds. Cryptographic primitives replace in place — Bulletproofs → Bulletproofs+ shipped this way, same Pedersen, same kernel sig, ~ 14% smaller witness. New Bitcoin primitives add cryptographic *options* alongside existing economic ones rather than displacing them: covenants enable fractional cBTC.zk alongside cBTC.tac, so users can choose either trust model. TAC’s structural roles persist and compound as the protocol surface grows — bond layer for workers, weight in governance, denominator and unit of account in canonical AMM pools (with optional protocol-fee accrual), reward asset for yield farms, and a confidential native asset in its own right (shielded amounts and addresses, mixer-composable, privkey-recoverable). New surfaces — additional pool families, stablecoin variants, channel-layer service economies — broaden the role set without changing what already ships. The indexer-validated meta-protocol pattern is a substrate that carries more powerful primitives as Bitcoin gains them, without ever asking users to migrate value out and back in.

15 Conclusion

We have proposed a self-custodial DeFi layer on Bitcoin L1, built from a composition of primitives. Two Groth16 circuit families — anonymous unique-spend (`withdraw.circum`) and amount confidentiality (AMM) — compose across every privacy-bearing surface. The indexer-validated meta-protocol pattern that already underwrites Runes and Ordinals is extended into a **collateral substrate**: TAC, a fixed 21M-base-unit asset, bonds workers against equivocation and backs fungible cBTC.tac; cBTC.zk’s “one leaf, two locks” construction gives trustless fixed-denomination BTC slots without a federation. Shielded amounts hide on-chain values, shielded addresses break recipient clustering at the output script, and the mixer breaks deposit–withdrawal linkage at the protocol-tree leaf — composed at the user’s choice of granularity, the three deliver full privacy across the tacit-asset surface. Merchants opt in only to shielded amounts and keep a public identity for accounting; payroll adds shielded addresses for unlinkable per-employee receipts; privacy-conscious users route through the mixer for unlinkability at every endpoint within the chosen anonymity set. Wallets reconstruct from the private key and Bitcoin chain alone.

The cryptographic claims hold under standard assumptions. The economic claims hold conditional on TAC market depth and an active liquidator population; limits are stated explicitly in §13. Every load-bearing property is either cryptographically enforced from chain alone, or bounded by an economic mechanism the protocol states in the open. Where Bitcoin already provides a primitive, tacit uses it. Where Bitcoin does not, tacit composes indexer-validated state and a fixed-cap bonded asset to fill the gap. As Bitcoin acquires more script primitives, the cryptographic surface widens; TAC continues to bond workers, weight governance, denominate canonical AMM pools (with protocol-fee accrual), reward yield farms, and carry its own confidential value — the structural roles compound as the protocol surface grows (§14).

Canonical specs: SPEC.md, AMM.md, MIXER.md, spec/CIRCUITS.md, spec/amendments/.